

MC Sharen Ganesh

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SUMMARY

Aspiring AI professional skilled in Python and machine learning frameworks. Experienced in applying AI tools like conversational agents and optimization models to solve real-world problems. Passionate about advancing expertise in neural networks, NLP, and AI research. Eager to contribute to cutting-edge AI innovations in academia and industry.

EDUCATION

2022 - Present	Diploma in Programming and Data Science at IIT Madras	(CGPA: 8.5)
2019 - 2023	B.Tech (EEE) at Bharati Vidyapeeth's College of Engineering	(CGPA: 8.9)
2019	Class 12th - CBSE	(86.17%)
2017	Class 10th - CBSE	(CGPA: 9.2)

WORK EXPERIENCE

GET, Bennet Coleman & Co (Times of India)

July 2023 - July 2024

- **Led IIoT system audit for Industry 4.0 adoption.** Performed **Exploratory Data Analysis (EDA)** on machine sensor data to identify operational inefficiency patterns, and collaborated with maintenance teams to create **predictive maintenance strategies**. Developed a user-friendly application to streamline life-enhancement activity scheduling.
- Created user-friendly machine parameter documentation, improving maintenance efficiency and reducing downtime.
- Contributed to Regioman Machine Upgradation Project by **modifying PLC code** for location-specific needs.

PROJECTS

Research Paper Abstraction Engine

Developed an AI-powered tool to accelerate academic research by dynamically parsing scientific papers and generating summaries across **9 abstraction levels** — from key-sentence extraction to high-level conceptual synthesis — using **Google's Gemini API** and **PyPDF2** for PDF processing. Designed **rule-based text segmentation** algorithms and integrated an interactive **Streamlit** dashboard with abstraction-level sliders, **reducing literature review time**.

AI-Based Teaching Assistant (Socratic Method)

Developed an **intelligent educational agent** capable of context-aware conversations by applying **Socratic dialogue** – a method of learning through questioning – which promotes deeper understanding and active recall, which are particularly effective for language acquisition. This project was developed using **Google's Gemini API**. **Chat history management**, **content summarization**, and **lesson categorization** were implemented in a streamlined **Streamlit** Interface.

SKILLS

Technical	Python • Machine Learning • LangChain • OpenCV • SQL • Bash • Solidity • ReactJS • HuggingFace PyTorch • Microsoft Azure • Neo4j • Database Management
Non-Technical	Research • Problem Solving • Documentation • Analytical skills • Indian Music

PUBLICATIONS

Study of Optimization Techniques for Optimal Location of FACTS Devices

<https://doi.org/10.1080/02522667.2022.2039470>

Developed **MATLAB** code for ascertaining the optimal sizing and location of **G-UPFCs** using Ant Colony Optimization in IEEE-14 bus systems. Presented in the **3rd International Conference on "Renewable Energy Potential for Sustainable Initiatives" (REPSI-2022, JIOS)**.

Chlorella pituita for an Obesity Free World; by Regulating Obesogen Activity

ISEF Project ID: 16543

Analyzed mice fibroblast cell line data (3T3-L1) using to assess the effect of **Polychlorinated Biphenyls (PCBs)** on obesity. Leveraged **statistical models** to evaluate the efficacy of *Chlorella pituita*, a freshwater alga, in countering obesogen-induced obesity. **Finalist** at the **International Science and Engineering Fair (ISEF)- 2018**.